Kubernetes - Storage and Secrets

Persistent volumes (PV):

Create a pv.yml file

```yaml
pv.yml U
datascientest-data-engineering > kub
apiVersion: v1
kind: PersistentVolume
metadata:
  name: datascientest-pv
spec:
  capacity:
    storage: 10Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: /mnt/data
  storageClassName: slow
```

Apply this configuration to create persistent volume, then check persistent volume creation

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f pv.yml
persistentvolume/datascientest-pv created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pv
NAME               CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS      CLAIM   STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
datascientest-pv   10Gi       RWO            Retain           Available           slow           <unset>                          7s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Dynamic PVC via StorageClass:

Check your available StorageClasses

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get storageclass
NAME                   PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
local-path (default)   rancher.io/local-path   Delete          WaitForFirstConsumer   false                  47h
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create a pvc.yml file under datascientest-pv

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ mkdir datascientest-pv && cd datascientest-pv
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$
```

```yaml
pvc.yml U
tascientest-data-engineering > kubernete
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: pvc-datascientest
spec:
  accessModes:
    - ReadWriteOnce
  storageClassName: local-path
  resources:
    requests:
      storage: 128Mi
```

Apply and inspect

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl apply -f pvc.yml
persistentvolumeclaim/pvc-datascientest created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl get pvc
NAME                STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
pvc-datascientest   Pending                                      local-path     <unset>                 22s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$
```

In datascientest-pv directory create Pod.yml file

pod.yml U ×

atascientest-data-engineering > kubernetes > datascientest-pv > ! pod.yml

```yaml
apiVersion: v1 # api version used to create Pod objects
kind: Pod # object type we want to create
metadata:
  name: datascientest-pod # Pod name
spec:
  containers:
    - name: nginx # container to create for Pod
      image: nginx # image to be used to create container
      imagePullPolicy: IfNotPresent # only if not present, download image
      volumeMounts:
        - name: datascientest-volume
          mountPath: /usr/share/nginx/html/ # directory in container to be mounted on volume
      ports:
        - containerPort: 80 # container listening port
  volumes:
    - name: datascientest-volume # volume name for persistent data storage
      persistentVolumeClaim:
        claimName: pvc-datascientest # PVC name used
```

Run it and verify

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl apply -f pod.yml
pod/datascientest-pod created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl get pod
NAME                                READY   STATUS              RESTARTS      AGE
curl                                1/1     Running             4 (54m ago)   22h
datascientest-pod                   0/1     ContainerCreating   0             8s
httpd-replicaset-67bgm              1/1     Running             3 (54m ago)   19h
httpd-replicaset-67nw5              1/1     Running             3 (54m ago)   19h
httpd-replicaset-6gnp9              1/1     Running             3 (54m ago)   19h
httpd-replicaset-hm7lz              1/1     Running             3 (54m ago)   19h
httpd-replicaset-rm2tc              1/1     Running             3 (54m ago)   19h
nginx-deployment-6476b5cb9d-fvhhl   1/1     Running             3 (54m ago)   23h
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$
```

List persistent volumes

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl get pv
NAME                                       CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS      CLAIM                       STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
datascientest-pv                           10Gi       RWO            Retain           Available                               slow           <unset>                          52m
pvc-1fbfc569-755c-436b-9f7f-d18c0407d8fc   128Mi      RWO            Delete           Bound       default/pvc-datascientest   local-path     <unset>                          3m13s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$
```

Check our PVC

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl get pvc
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
pvc-datascientest   Bound    pvc-1fbfc569-755c-436b-9f7f-d18c0407d8fc   128Mi      RWO            local-path     <unset>                 13m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$
```

Check that the Pod is active

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl get pod | grep datascientest-pod
datascientest-pod                   1/1     Running   0             7m31s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$
```

Test persistence / overwrite index.html inside the Pod

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl exec -it datascientest-pod -- /bin/bash
root@datascientest-pod:/# echo "DATASCIENTEST" > /usr/share/nginx/html/index.html
root@datascientest-pod:/# cat /usr/share/nginx/html/index.html
DATASCIENTEST
root@datascientest-pod:/# exit
exit
```

Then delete and recreate the Pod as before, “DATASCIENTEST” should still be visible

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl delete pod datascientest-pod
pod "datascientest-pod" deleted
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl apply  -f pod.yml
pod/datascientest-pod created
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes/datascientest-pv$ kubectl exec -it datascientest-pod -- cat /usr/share/nginx/html/index.html
DATASCIENTEST
```

Secrets:
Select password and convert to base64 to create secret containing MYSQL_ROOT_PASSWORD

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ echo -n 'Datascientest2023@!!!' | base64
RGF0YXNjaWVudGVzdDIwMjNAISEh
```

Create a secret-datascientest-mysql.yml file

```
! secret-datascientest-mysql.yml U ×
datascientest-data-engineering > kubernetes > ! secre
1   apiVersion: v1
2   kind: Secret
3   metadata:
4     name: mariadb-datascientest-rootpass
5   type: Opaque
6   data:
7     password: RGF0YXNjaWVudGVzdDIwMjNAISE=
```

Apply object config

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f secret-datascientest-mysql.yml
secret/mariadb-datascientest-rootpass created
```

Display list of secrets

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get secret
NAME                             TYPE     DATA   AGE
mariadb-datascientest-rootpass   Opaque   1      3m45s
```

Get more info about our secret with describe option

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe secret mariadb-datascientest-rootpass
Name:         mariadb-datascientest-rootpass
Namespace:    default
Labels:       <none>
Annotations:  <none>

Type:  Opaque

Data
====
password:  20 bytes
```

Decode our secret and create user-secret

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get secret mariadb-datascientest-rootpass -o jsonpath='{.data.password}' | base64 --decod
e -
Datascientest2023@!!ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create secret generic mariadb-user-datascientest --from-literal=MYSQL
_USER=datascientestuser --from-literal=MYSQL_PASSWORD=DatascientestMariadb@...
secret/mariadb-user-datascientest created
```

Check that username and password have been created and stored correctly

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get secret mariadb-user-datascientest -o jsonpath='{.data.MYSQL_USER}' | base64 --decode
-
datascientestuserubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get secret mariadb-user-datascientest -o jsonpath='{.data.MYSQL_PASSWORD
}' | base64 --decode -
DatascientestMariadb@...ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

ConfigMaps:

Create max_allowed_packet.cnf file

```
max_allowed_packet.cnf U
datascientest-data-engineering > kube
1   [mysqld]
2   max_allowed_packet = 128M
```

Create mariadb-datascientest-config ConfigMap from prior file

```
DatascientestMariadb@...ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create configmap mariadb-datascientest-config --from-file=max_all
owed_packet.cnf
configmap/mariadb-datascientest-config created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

View ConfigMap contents with kubectl describe command

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe cm mariadb-datascientest-config
Name:         mariadb-datascientest-config
Namespace:    default
Labels:       <none>
Annotations:  <none>

Data
====
max_allowed_packet.cnf:
----
[mysqld]
max_allowed_packet = 128M


BinaryData
====

Events:  <none>
```

Edit to have max_allowed_packet set to 256M

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl edit configmap mariadb-datascientest-config
configmap/mariadb-datascientest-config edited
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe cm mariadb-datascientest-config
Name:         mariadb-datascientest-config
Namespace:    default
Labels:       <none>
Annotations:  <none>

Data
====
max_allowed_packet.cnf:
----
[mysqld]
max_allowed_packet = 256M


BinaryData
====

Events:  <none>
```

Check Configmap update

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ #Note that the '.' in max_allowed_packet.cnf must be preceeded by a \
datascientest-config -o "jsonpath={.data['max_alloubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get configmap mariadb-datascientest-con
fig -o "jsonpath={.data['max_allowed_packet\.cnf']}" #
[mysqld]
max_allowed_packet = 256M
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create a MariaDB instance:

Create statefulset.yml file

```
statefulset.yml U ●
datascientest-data-engineering > kubernetes > statefulset.yml
```

```yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mariadb-deployment
  labels:
    app: mariadb
spec:
  serviceName: "maria"
  replicas: 1
  selector:
    matchLabels:
      app: mariadb
  template:
    metadata:
      labels:
        app: mariadb
    spec:
      containers:
        - name: mariadb
          image: docker.io/mariadb:10.4
          ports:
            - containerPort: 3306
              protocol: TCP

          # ---- mount Secrets as env vars ----
          env:
            - name: MYSQL_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mariadb-datascientest-rootpass
                  key: password
          envFrom:
            - secretRef:
                name: mariadb-user-datascientest

          # ---- mount volumes into the container ----
          volumeMounts:
            # persistent data
            - name: mariadb-volume-1
              mountPath: /var/lib/mysql
            # custom config from ConfigMap
            - name: mariadb-datascientest-config-volume
              mountPath: /etc/mysql/conf.d

      # ---- define volumes ----
      volumes:
        - name: mariadb-volume-1
          emptyDir: {}
        # pull in max_allowed_packet.cnf from ConfigMap
        - name: mariadb-datascientest-config-volume
          configMap:
            name: mariadb-datascientest-config
            items:
              - key: max_allowed_packet.cnf
                path: max_allowed_packet.cnf
```

Apply config and verify Pod comes up

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f statefulset.yml
statefulset.apps/mariadb-deployment created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pods -l app=mariadb
NAME                   READY   STATUS              RESTARTS   AGE
mariadb-deployment-0   0/1     ContainerCreating   0          11s
```

Check that Secrets landed as environment variables

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl exec -it mariadb-deployment-0 -- env | grep MYSQL
MYSQL_PASSWORD=DatascientestMariadb@...
MYSQL_USER=datascientestuser
MYSQL_ROOT_PASSWORD=Datascientest2023@!!
```

Ensure ConfigMap file is in place

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl exec -it mariadb-deployment-0 -- ls /etc/mysql/conf.d
max_allowed_packet.cnf
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl exec -it mariadb-deployment-0 -- \
>   cat /etc/mysql/conf.d/max_allowed_packet.cnf
[mysqld]
max_allowed_packet = 256M
```

Login and confirm MariaDB reads both

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl exec -it mariadb-deployment-0 -- /bin/sh
# mysql -uroot -p${MYSQL_ROOT_PASSWORD} -e 'show databases;'
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
# mysql -uroot -p${MYSQL_ROOT_PASSWORD} \
  -e "SHOW>  VARIABLES LIKE 'max_allowed_packet';"
+--------------------+-----------+
| Variable_name      | Value     |
+--------------------+-----------+
| max_allowed_packet | 268435456 |
+--------------------+-----------+
# exit
```

Health Checks:

Create vote-deploy-probes.yml file

! vote-deploy-probes.yaml U ×

datascientest-data-engineering > kubernetes > ! vote-deploy-probes.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vote
spec:
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 2
      maxUnavailable: 1
  revisionHistoryLimit: 4
  replicas: 12
  minReadySeconds: 20
  selector:
    matchLabels:
      role: voting
    matchExpressions:
      - { key: version, operator: In, values: [v1,v2,v3,v4,v5] }
  template:
    metadata:
      name: vote
      labels:
        app: python
        role: voting
        version: v1
    spec:
      containers:
      - name: app
        image: schoolofdevops/vote:v1
        resources:
          requests:
            memory: "64Mi"
            cpu: "50m"
          limits:
            memory: "128Mi"
            cpu: "250m"
        # —— Liveness Probe ————————————————
        livenessProbe:
          tcpSocket:
            port: 80
          initialDelaySeconds: 5
          periodSeconds: 5
        # —— Readiness Probe ———————————————
        readinessProbe:
          httpGet:
            path: /
            port: 80
          initialDelaySeconds: 5
          periodSeconds: 3
```

Deploy it

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f vote-deploy-probes.yaml
deployment.apps/vote created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Verify by listing pods created

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe deploy vote
Name:                   vote
Namespace:              default
CreationTimestamp:      Wed, 02 Jul 2025 12:47:37 +0000
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               role=voting,version in (v1,v2,v3,v4,v5)
Replicas:               12 desired | 12 updated | 12 total | 5 available | 7 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        20
RollingUpdateStrategy:  1 max unavailable, 2 max surge
Pod Template:
  Labels:  app=python
           role=voting
           version=v1
  Containers:
   app:
    Image:      schoolofdevops/vote:v1
    Port:       <none>
    Host Port:  <none>
    Limits:
      cpu:     250m
      memory:  128Mi
    Requests:
      cpu:         50m
      memory:      64Mi
    Liveness:      tcp-socket :80 delay=5s timeout=1s period=5s #success=1 #failure=3
    Readiness:     http-get http://:80/ delay=5s timeout=1s period=3s #success=1 #failure=3
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      False   MinimumReplicasUnavailable
  Progressing    True    ReplicaSetUpdated
OldReplicaSets:  <none>
NewReplicaSet:   vote-6dbf5d9fbc (12/12 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  48s   deployment-controller  Scaled up replica set vote-6dbf5d9fbc from 0 to 12
```

Force a liveness failure (livenessProbe)

Edit port from 80 to 8888 for example

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl edit deploy vote
deployment.apps/vote edited
```

Observe liveness failures

Display Pod list

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pods
NAME                                READY   STATUS    RESTARTS       AGE
curl                                1/1     Running   4 (3h15m ago)  24h
datascientest-pod                   1/1     Running   0              118m
httpd-replicaset-67bgm              1/1     Running   3 (3h15m ago)  21h
httpd-replicaset-67nw5              1/1     Running   3 (3h15m ago)  21h
httpd-replicaset-6gnp9              1/1     Running   3 (3h15m ago)  21h
httpd-replicaset-hm7lz              1/1     Running   3 (3h15m ago)  21h
httpd-replicaset-rm2tc              1/1     Running   3 (3h15m ago)  21h
mariadb-deployment-0                1/1     Running   0              24m
nginx-deployment-6476b5cb9d-fvhhl   1/1     Running   3 (3h15m ago)  25h
vote-6dbf5d9fbc-5kf4w               1/1     Running   0              9m58s
vote-6dbf5d9fbc-68snc               1/1     Running   0              9m58s
vote-6dbf5d9fbc-9hfbq               1/1     Running   0              9m58s
vote-6dbf5d9fbc-dt48t               1/1     Running   0              9m58s
vote-6dbf5d9fbc-fspff               1/1     Running   0              9m58s
vote-6dbf5d9fbc-gq4sr               1/1     Running   0              9m58s
vote-6dbf5d9fbc-gzxzk               1/1     Running   0              9m58s
vote-6dbf5d9fbc-j55zm               1/1     Running   0              9m58s
vote-6dbf5d9fbc-pbzqt               1/1     Running   0              9m58s
vote-6dbf5d9fbc-prm4t               1/1     Running   0              9m58s
vote-6dbf5d9fbc-v8qkk               1/1     Running   0              9m58s
vote-76d4c569d7-2ltsw               1/1     Running   4 (17s ago)    98s
vote-76d4c569d7-8kcrm               1/1     Running   4 (17s ago)    98s
vote-76d4c569d7-wdgrx               1/1     Running   4 (16s ago)    98s
```

Describe a newly created Pod

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe Pod vote-76d4c569d7-2ltsw
Name:             vote-76d4c569d7-2ltsw
Namespace:        default
Priority:         0
Service Account:  default
Node:             ip-172-31-18-113/172.31.18.113
Start Time:       Wed, 02 Jul 2025 12:55:57 +0000
Labels:           app=python
                  pod-template-hash=76d4c569d7
                  role=voting
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.42.0.116
IPs:
  IP:           10.42.0.116
Controlled By:  ReplicaSet/vote-76d4c569d7
Containers:
  app:
    Container ID:   containerd://29c8b4753162ad0b6e01d854fb15366b3802a9b3d1d5f0ec3b29eb807e238bd9
    Image:          schoolofdevops/vote:v1
    Image ID:       docker.io/schoolofdevops/vote@sha256:0bcdeb55763d0e535f287191e7d000009a4540950aed864bfb0671abaf596e0d
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       CrashLoopBackOff
    Last State:     Terminated
      Reason:       Completed
      Exit Code:    0
      Started:      Wed, 02 Jul 2025 12:57:19 +0000
      Finished:     Wed, 02 Jul 2025 12:57:38 +0000
    Ready:          False
    Restart Count:  4
    Limits:
      cpu:     250m
      memory:  128Mi
    Requests:
      cpu:        50m
      memory:     64Mi
    Liveness:     tcp-socket :8888 delay=5s timeout=1s period=5s #success=1 #failure=3
    Readiness:    http-get http://:80/ delay=5s timeout=1s period=3s #success=1 #failure=3
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-zxvcm (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-zxvcm:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   Burstable
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
```

```
Events:
  Type     Reason     Age                 From               Message
  ----     ------     ----                ----               -------
  Normal   Scheduled  2m1s                default-scheduler  Successfully assigned default/vote-76d4c569d7-2ltsw to ip-172-31-18-113
  Normal   Pulled     40s (x5 over 119s)  kubelet            Container image "schoolofdevops/vote:v1" already present on machine
  Normal   Created    40s (x5 over 119s)  kubelet            Created container: app
  Normal   Started    40s (x5 over 118s)  kubelet            Started container app
  Warning  Unhealthy  39s (x3 over 81s)   kubelet            Readiness probe failed: Get "http://10.42.0.116:80/": dial tcp 10.42.0.116:80: connect: connection refused
  Warning  Unhealthy  21s (x15 over 111s) kubelet            Liveness probe failed: dial tcp 10.42.0.116:8888: connect: connection refused
  Normal   Killing    21s (x5 over 101s)  kubelet            Container app failed liveness probe, will be restarted
  Warning  BackOff    6s (x4 over 21s)    kubelet            Back-off restarting failed container app in pod vote-76d4c569d7-2ltsw_default(2d627dd5-76da-4c87-9366-37962bd2
a5e3)
```

Fix broken livenessProbe by restoring port from 8888 to 80 and wait for Pods to recover

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl edit deploy vote
deployment.apps/vote edited
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Display information on Kubernetes cluster deployments

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get deploy,rs,pods
NAME                              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-deployment   1/1     1            1           25h
deployment.apps/vote               12/12   12           12          16m

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/httpd-replicaset              5         5         5       21h
replicaset.apps/nginx-deployment-6476b5cb9d   1         1         1       25h
replicaset.apps/vote-6dbf5d9fbc               12        12        12      16m
replicaset.apps/vote-76d4c569d7               0         0         0       7m58s

NAME                                    READY   STATUS    RESTARTS         AGE
pod/curl                                1/1     Running   4 (3h21m ago)    25h
pod/datascientest-pod                   1/1     Running   0                125m
pod/httpd-replicaset-67bgm              1/1     Running   3 (3h21m ago)    21h
pod/httpd-replicaset-67nw5              1/1     Running   3 (3h21m ago)    21h
pod/httpd-replicaset-6gnp9              1/1     Running   3 (3h21m ago)    21h
pod/httpd-replicaset-hm7lz              1/1     Running   3 (3h21m ago)    21h
pod/httpd-replicaset-rm2tc              1/1     Running   3 (3h21m ago)    21h
pod/mariadb-deployment-0                1/1     Running   0                31m
pod/nginx-deployment-6476b5cb9d-fvhhl   1/1     Running   3 (3h21m ago)    25h
pod/vote-6dbf5d9fbc-5kf4w               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-68snc               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-9hfbq               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-dkd8k               1/1     Running   0                115s
pod/vote-6dbf5d9fbc-dt48t               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-fspff               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-gq4sr               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-gzxzk               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-j55zm               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-pbzqt               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-prm4t               1/1     Running   0                16m
pod/vote-6dbf5d9fbc-v8qkk               1/1     Running   0                16m
```

Describe Pod vote-6dbf5d9fbc-v8qkk launched after changing preparation probe

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe Pod vote-6dbf5d9fbc-v8qkk
Name:             vote-6dbf5d9fbc-v8qkk
Namespace:        default
Priority:         0
Service Account:  default
Node:             ip-172-31-18-113/172.31.18.113
Start Time:       Wed, 02 Jul 2025 12:47:37 +0000
Labels:           app=python
                  pod-template-hash=6dbf5d9fbc
                  role=voting
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.42.0.114
IPs:
  IP:           10.42.0.114
Controlled By:  ReplicaSet/vote-6dbf5d9fbc
Containers:
  app:
    Container ID:   containerd://caa1a95ae4459904311e41e9da0c64ea6c011fba4284349ff3856b761e46df1f
    Image:          schoolofdevops/vote:v1
    Image ID:       docker.io/schoolofdevops/vote@sha256:0bcdeb55763d0e535f287191e7d000009a4540950aed864bfb0671abaf596e0d
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Wed, 02 Jul 2025 12:47:50 +0000
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     250m
      memory:  128Mi
    Requests:
      cpu:        50m
      memory:     64Mi
    Liveness:     tcp-socket :80 delay=5s timeout=1s period=5s #success=1 #failure=3
    Readiness:    http-get http://:80/ delay=5s timeout=1s period=3s #success=1 #failure=3
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-xp78h (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-xp78h:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   Burstable
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
```

```
Events:
  Type     Reason     Age   From               Message
  ----     ------     ----  ----               -------
  Normal   Scheduled  17m   default-scheduler  Successfully assigned default/vote-6dbf5d9fbc-v8qkk to ip-172-31-18-113
  Normal   Pulling    17m   kubelet            Pulling image "schoolofdevops/vote:v1"
  Normal   Pulled     17m   kubelet            Successfully pulled image "schoolofdevops/vote:v1" in 5.778s (5.778s including waiting). Image size: 29048424 bytes.
  Normal   Created    17m   kubelet            Created container: app
  Normal   Started    17m   kubelet            Started container app
  Warning  Unhealthy  17m   kubelet            Readiness probe failed: Get "http://10.42.0.114:80/": context deadline exceeded (Client.Timeout exceeded while awaiting heade
rs)
```